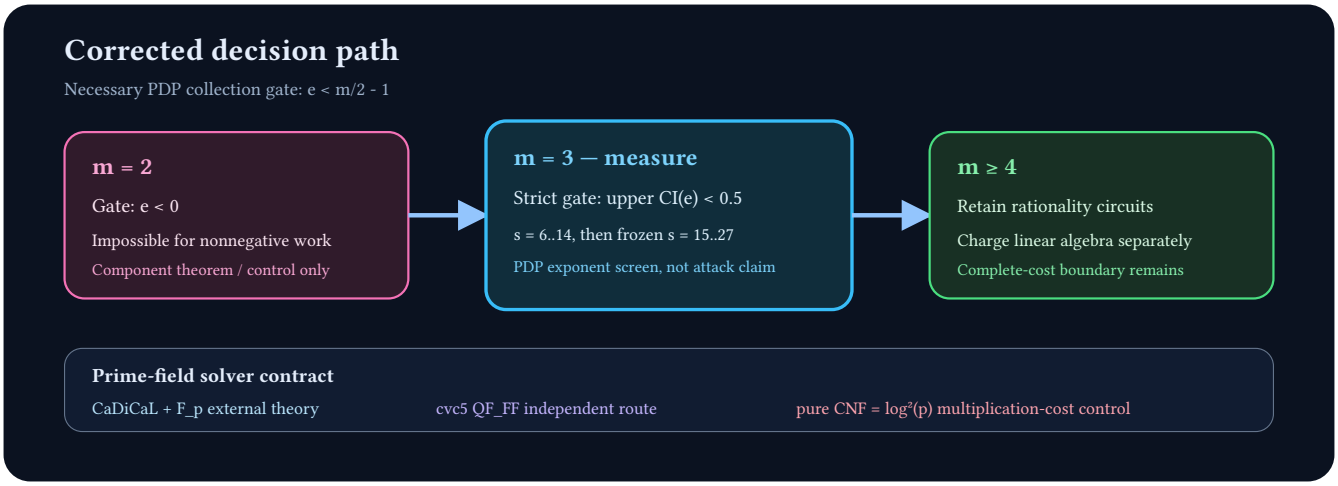
Prime-field index calculus: arity and solver correction

DEC-20261031-b51282



The programme now treats every accepted arity-two theorem cell as component evidence, not as an exponent-changing ECDLP result. With $s = \log_2 \frac{p}{m}$, PDP work 2^{es} and $\Theta(2^s)$ required relations, collection costs $p^{\frac{e+1}{m}}$ up to polynomial factors. Strict improvement on rho requires

$$e < \frac{m}{2} - 1.$$

At $m = 2$ this requires $e < 0$ and is impossible for nonnegative work. At $m = 3$, the first relevant PDP screen, the necessary threshold is $e < 0.5$. This is not sufficient for a complete index-calculus win: the retained PKM bridge still places its linear-algebra crossover at $m \geq 4$.

Correction boundaries

DEC-20261031-cd6460 remains the record of the standalone outer-chart metadata defect. DEC-20261031-ca1521 remains the fixed F5/F7 software evidence rejecting R1. This decision does not reopen either suite. It corrects the later inference: fixing the checker does not turn an $m = 2$ component theorem into an ECDLP exponent result.

Boolean quotient equations $z^2 - z$ avoid carry semantics in the mathematical field model, but a general bit-level F_p multiplication still costs $\Theta(\log^2 p)$ gates. The theorem-bearing candidates are CaDiCaL with an external F_p theory and independently replayed explanation clauses, plus a pinned cvc5 QF_FF export of the same predicate. Pure CNF reports encoding size and complete circuit cost only.

Retained predicate

The corrected $m = 3$ predicate uses three disjoint one-hot windows and two chained S_3 relations, never an expanded S_4. Its nonexceptional rational lift is

$$(2t\lambda - s_0)^2 = f(x),$$

with discriminant $16t^2 f(x)$; $t = 0$ remains an explicit exceptional branch. Under a rational non-2-torsion target, odd prime whole-group order and disjoint windows, the three-summand Frobenius argument supplies rationality. At arity four or higher explicit rationality circuits remain mandatory. Every out-of-domain field or index word receives a forbidden-word clause.

The Q1421 source bound to SHA-256 `0a0d2c673e9484f4b217917ee306e7663140e4d4618d8bb0ed8dc91947d08bc6` is an external-propagator precedent over F_{2^n} only. It is not the missing F_p theory.

Fixed next measurement

Cryptanalysis PR 1343 fixes $m = 3$, every integer $s = 6..14$, and a strict Holm-corrected upper-interval gate $e < 0.5$. Controls are exact two-window enumeration ($e = 2$), a curve-free support-matched null, planted and exact-UNSAT instances, fresh-process A/A, and pure-CNF size accounting. Complete work includes target-dependent construction, theory work, explanation replay and independent group-law verification.

The 40–80-bit path is $s = 13..27$, or about 39–81 field bits at $m = 3$. The initial ladder reaches $s = 13, 14$; $s = 15..27$ requires a separately frozen amendment. At 80 bits rho is about 2^{40} group operations, so this measures the PDP exponent rather than predicts a wall-clock win.

No F_p propagator, held-out instance, cvc5 benchmark, field census or exponent fit exists in this correction. Scientific runs: zero. Performance claims: zero.

Primary-source context: Petit, Kosters and Messeng, *Algebraic approaches for the elliptic curve discrete logarithm problem over prime fields*, PKC 2016, DOI 10.1007/978-3-662-49387-8_1. Exact identities: `report-inputs.json`.